Ogbonna Arinze

arinzeogbo@gmail.com | LinkedIn

SUMMARY

Full Stack Software Developer with 8 years of professional experience building scalable web and mobile-ready applications across healthcare, fintech, civic tech, and enterprise SaaS environments. Deep expertise in **NestJS** and **NodeJS** backend architecture, **React** and **Next.js** frontend systems, **TypeScript**, **GraphQL** APIs, and **PostgreSQL** databases. Proven track record delivering production-grade platforms with **Docker** containerization, CI/CD automation, real-time data streaming, and third-party service integrations, while contributing to technical decisions on architecture, development priorities, and engineering standards in agile team environments.

CORE COMPETENCIES

- **Backend & APIs:** NestJS, NodeJS, REST APIs, GraphQL, JWT authentication, RBAC, SSE streaming, and third-party service integrations.
- **Frontend & Mobile:** React, Next.js, Vue.js, TypeScript, Redux, Zustand, Tailwind CSS, Radix UI, and responsive cross-device layouts.
- **Databases & ORM:** PostgreSQL (pgvector), MongoDB, Prisma, database schema design, and data modeling.
- **DevOps & Cloud:** Docker, Docker Compose, GitHub Actions, CI/CD pipelines, AWS (CloudFront, S3), Heroku, Traefik, and Nx monorepo orchestration.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Mastra, vector search (pgvector, Chroma), embeddings, and multi-agent workflows.
- **Architecture & Leadership:** Scalable system design, monorepos (Nx, pnpm), architecture reviews, mentorship, Agile/Scrum, and security best practices.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension** in **TypeScript**, optimizing workflows for clinical site integrations and enhancing adoption across multiple teams.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query their data in real time via a conversational interface.
- Implemented **JWT-based authentication** and **role-based access control** on the frontend, securely handling access and refresh tokens with HTTP-only cookies and collaborating with backend engineers to enforce permission-based UI rendering.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Optimized application performance by implementing **lazy loading, code splitting**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports and more reliable releases.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices with precise user story planning.

- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, processing thousands to millions of daily transactions and boosting settlement rates by **80%** through optimized transaction processing workflows.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React**, **Redux**, **RxJS** for real-time data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface for live transaction monitoring.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs**, streamlining user role and permission management with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** to power new features, translating business requirements into efficient **MongoDB** database models and schemas with a focus on testable, maintainable code.
- Transformed user journeys and Figma designs into reusable **React** components, ensuring a consistent and intuitive user interface across the application.
- Engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and structured task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing all functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app (22 interactive chart types with Recharts), Next.js admin dashboard, **NestJS** document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with **pnpm** workspaces.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and wrote custom migration tooling that filters out Mastra-managed chunk tables from Prisma's drift detection.
- Built a **multi-agent AI system** using the **Mastra** framework with 5 specialized agents and a three-stage intent classification pipeline (keyword scoring, LLM classification with caching, inferred fallback), powered by a **hybrid RAG pipeline** combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats to prevent reverse proxy timeouts, phone OTP and Telegram OAuth authentication, and shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and **Traefik** reverse proxy routing.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*